

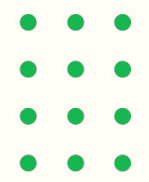
Capstone: Understanding

Customer's Credit Data

Group Members:

- Ivan Richmond Jumawan
- Ammiel Majan
- Isaiah Lex Javier
- Adan Hernandez
- Owen Mallapre
- Edson Broderick Lim

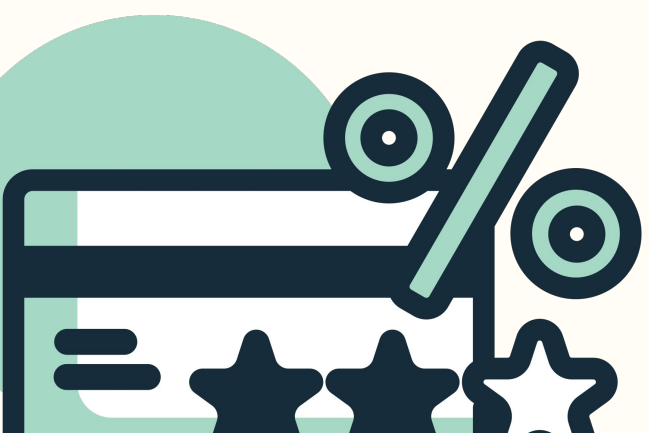




Questions



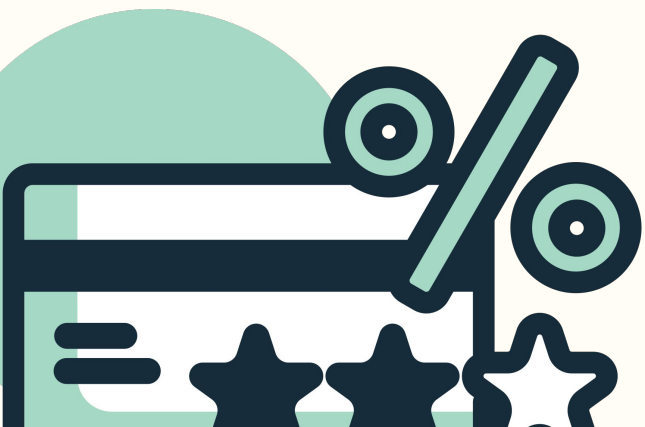
1. Can we divide the data into clusters?
2. What optimal number of clusters can the data be divided into?
3. What are the defining features of each cluster?
 - a. Can we infer the buying strategies of each cluster?
4. Can we use this data to help streamline the approval process of credit card application?





Description of Features

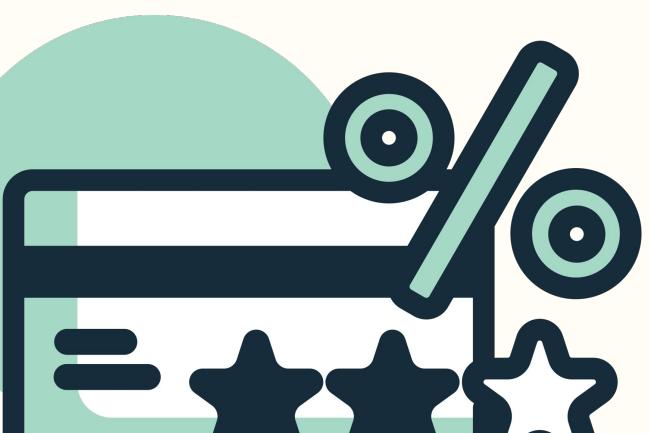
- **CUST_ID** - Unique identifier for each Credit Card holder
- **BALANCE** - Balance amount left to make purchases
- **BALANCE_FREQUENCY** - How frequently the balance is updated (1 = frequent, 0 = infrequent)
- **PURCHASES** - Total amount of purchases
- **ONEOFF_PURCHASES** - Largest single purchase
- **INSTALLMENTS_PURCHASES** - Total amount of purchases made on an installment plan
- **CASH_ADVANCE** - Cash in advance given
- **PURCHASES_FREQUENCY** - How frequently purchases are made (1 = frequent, 0 = infrequent)
- **ONEOFFPURCHASESFREQUENCY** - How frequently one-time purchases are made (1 = frequent, 0 = infrequent)
- **PURCHASESINSTALLMENTSFREQUENCY** - How frequently purchases in installments are made (1 = frequent, 0 = infrequent)
- **CASHADVANCEFREQUENCY** - How frequently the cash in advance being paid
- **CASHADVANCETRX** - Number of transactions made with "Cash in Advanced"
- **PURCHASES_TRX** - Number of purchase transactions made
- **CREDIT_LIMIT** - Maximum amount of credit
- **PAYMENTS** - Amount of payment done
- **MINIMUM_PAYMENTS** - Minimum amount of payments made
- **PRCFULLPAYMENT** - Percent of full payment paid

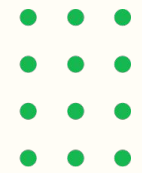


Properties of the Dataframe

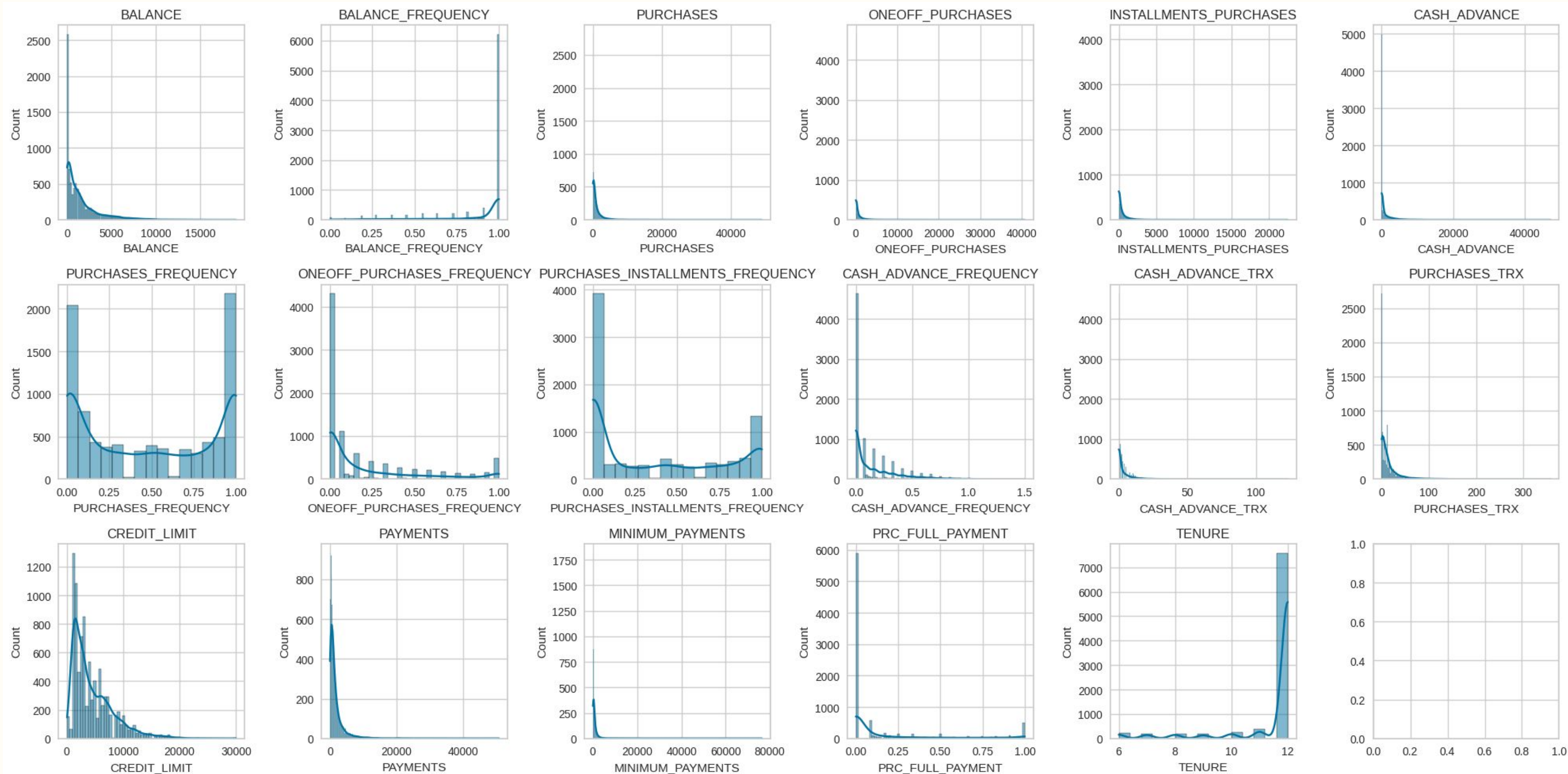
- Dimensions - (8950,19)

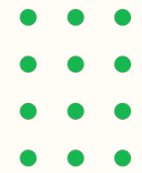
```
Data columns (total 18 columns):  
#      Column                                     Non-Null Count  Dtype  
---  -  
0     CUST_ID                                     8950 non-null   object  
1     BALANCE                                     8950 non-null   float64  
2     BALANCE_FREQUENCY                         8950 non-null   float64  
3     PURCHASES                                  8950 non-null   float64  
4     ONEOFF_PURCHASES                         8950 non-null   float64  
5     INSTALLMENTS_PURCHASES                   8950 non-null   float64  
6     CASH_ADVANCE                              8950 non-null   float64  
7     PURCHASES_FREQUENCY                      8950 non-null   float64  
8     ONEOFF_PURCHASES_FREQUENCY              8950 non-null   float64  
9     PURCHASES_INSTALLMENTS_FREQUENCY        8950 non-null   float64  
10    CASH_ADVANCE_FREQUENCY                   8950 non-null   float64  
11    CASH_ADVANCE_TRX                        8950 non-null   int64  
12    PURCHASES_TRX                          8950 non-null   int64  
13    CREDIT_LIMIT                            8949 non-null   float64  
14    PAYMENTS                                8950 non-null   float64  
15    MINIMUM_PAYMENTS                        8637 non-null   float64  
16    PRC_FULL_PAYMENT                        8950 non-null   float64  
17    TENURE                                  8950 non-null   int64  
dtypes: float64(14), int64(3), object(1)
```



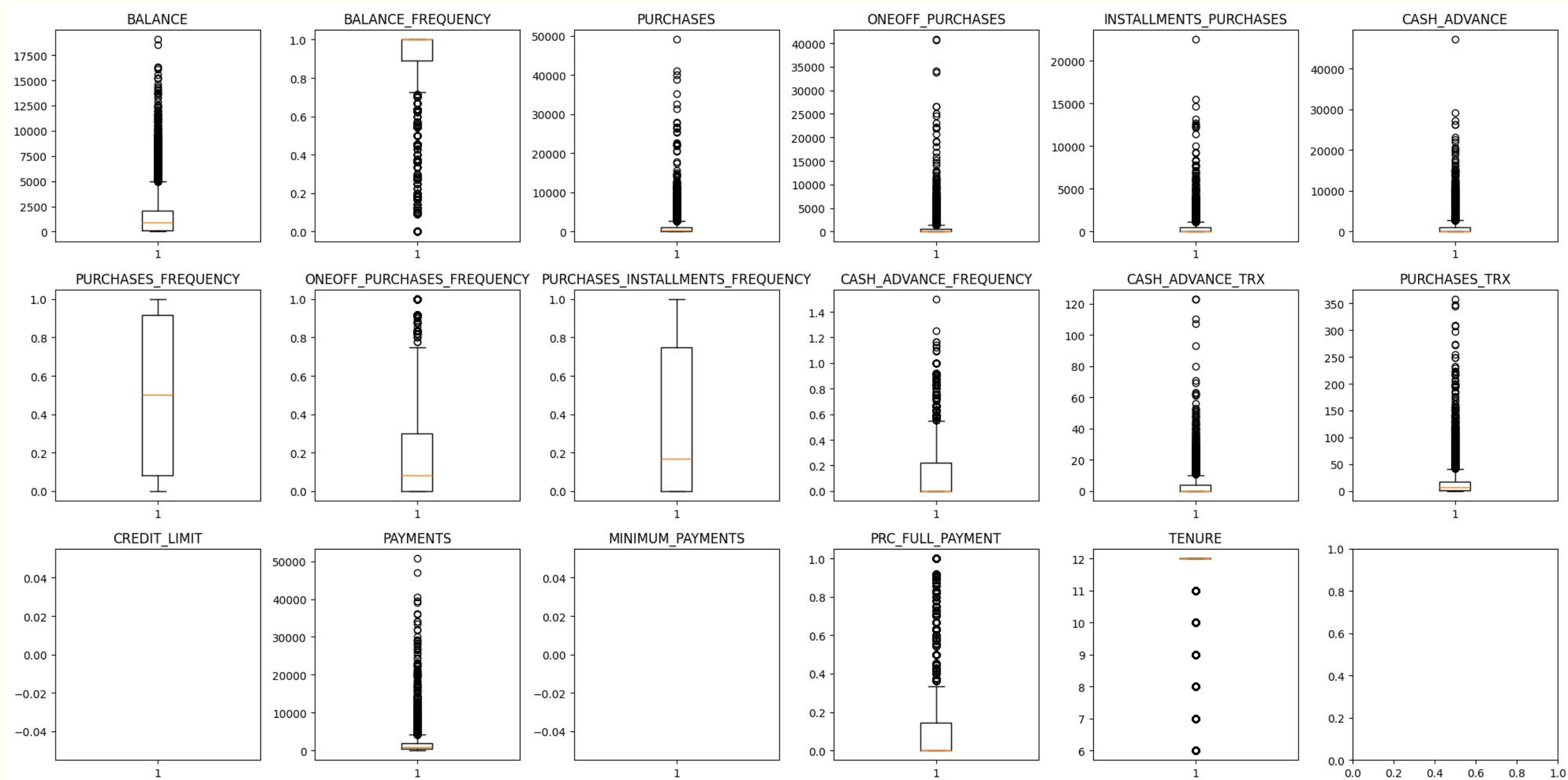


Histogram





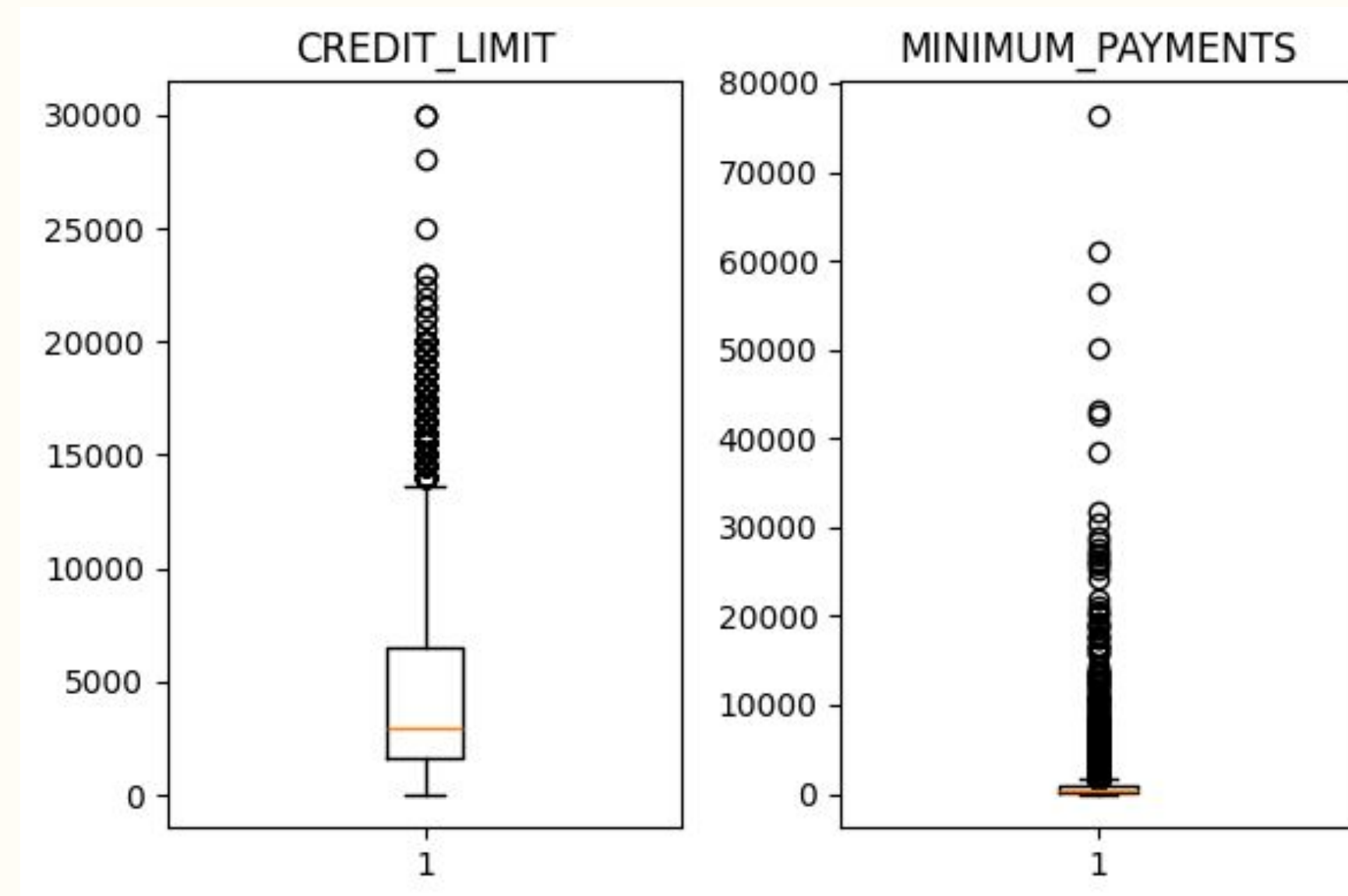
Box Plots



Missing Values and corrected box plot

Checking the data for missing values:

	0
CUST_ID	0
BALANCE	0
BALANCE_FREQUENCY	0
PURCHASES	0
ONEOFF_PURCHASES	0
INSTALLMENTS_PURCHASES	0
CASH_ADVANCE	0
PURCHASES_FREQUENCY	0
ONEOFF_PURCHASES_FREQUENCY	0
PURCHASES_INSTALLMENTS_FREQUENCY	0
CASH_ADVANCE_FREQUENCY	0
CASH_ADVANCE_TRX	0
PURCHASES_TRX	0
CREDIT_LIMIT	1
PAYMENTS	0
MINIMUM_PAYMENTS	313
PRC_FULL_PAYMENT	0
TENURE	0



Preprocessing

```
df['AVG_TRX_AMOUNT'] = df['PURCHASES'] / df['PURCHASES_TRX']  
df.fillna({'AVG_TRX_AMOUNT': 0}, inplace=True)
```

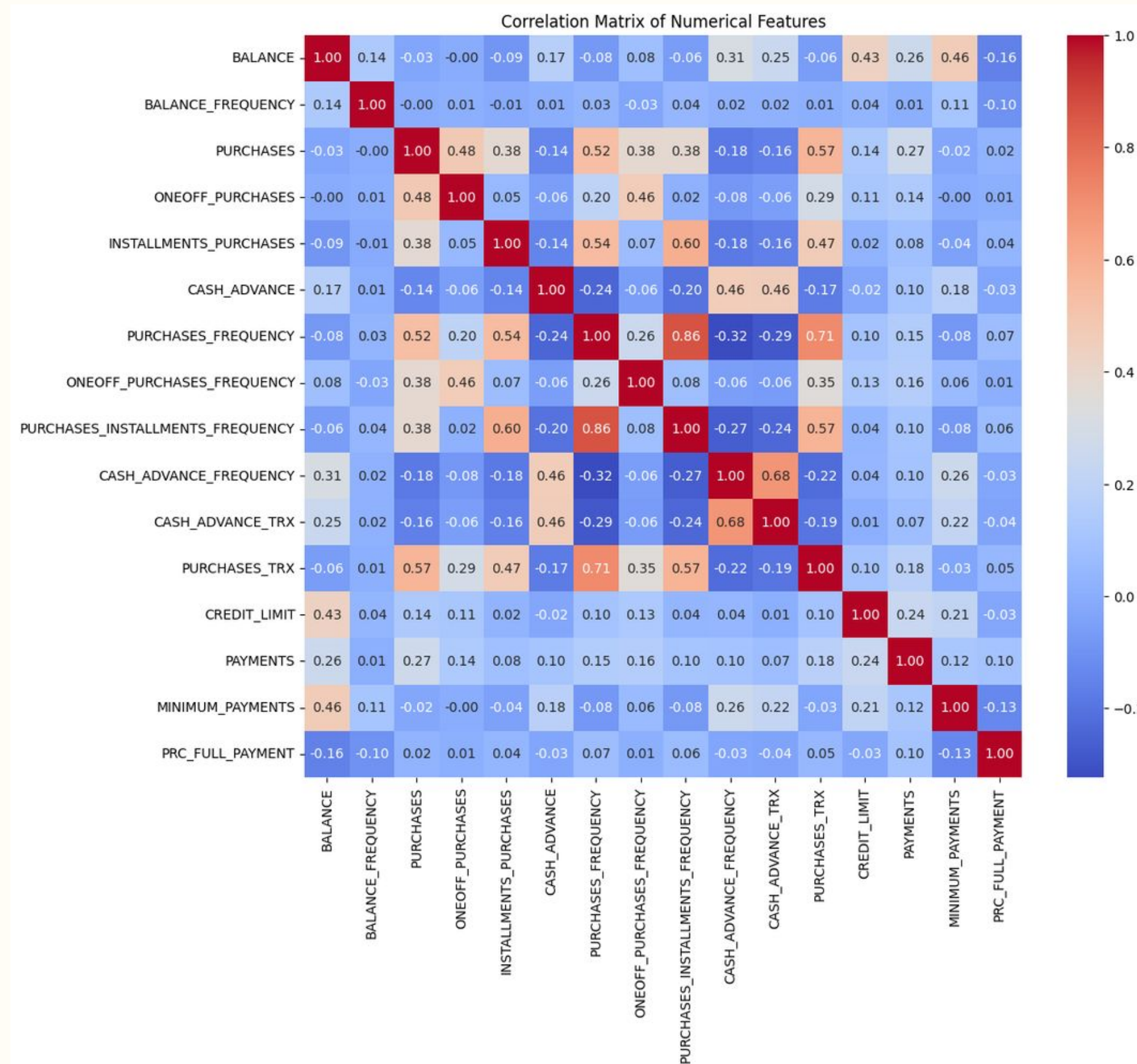
Replace NaN with 0's. These are rows wherein there were no purchases and/or purchase transactions. It is possible that they got a "free item," which would result to a 0 PURCHASE but a >0 PURCHASE_TRX. It is right to change this to 0.

```
[21] infinite_indices = df[np.isinf(df['AVG_TRX_AMOUNT'])].index  
print("Indexes with infinite values in 'AVG_TRX_AMOUNT':", infinite_indices)  
  
df['AVG_TRX_AMOUNT'] = df['AVG_TRX_AMOUNT'].replace([np.inf, -np.inf], np.nan)  
df.dropna(subset=['AVG_TRX_AMOUNT'], inplace=True)
```

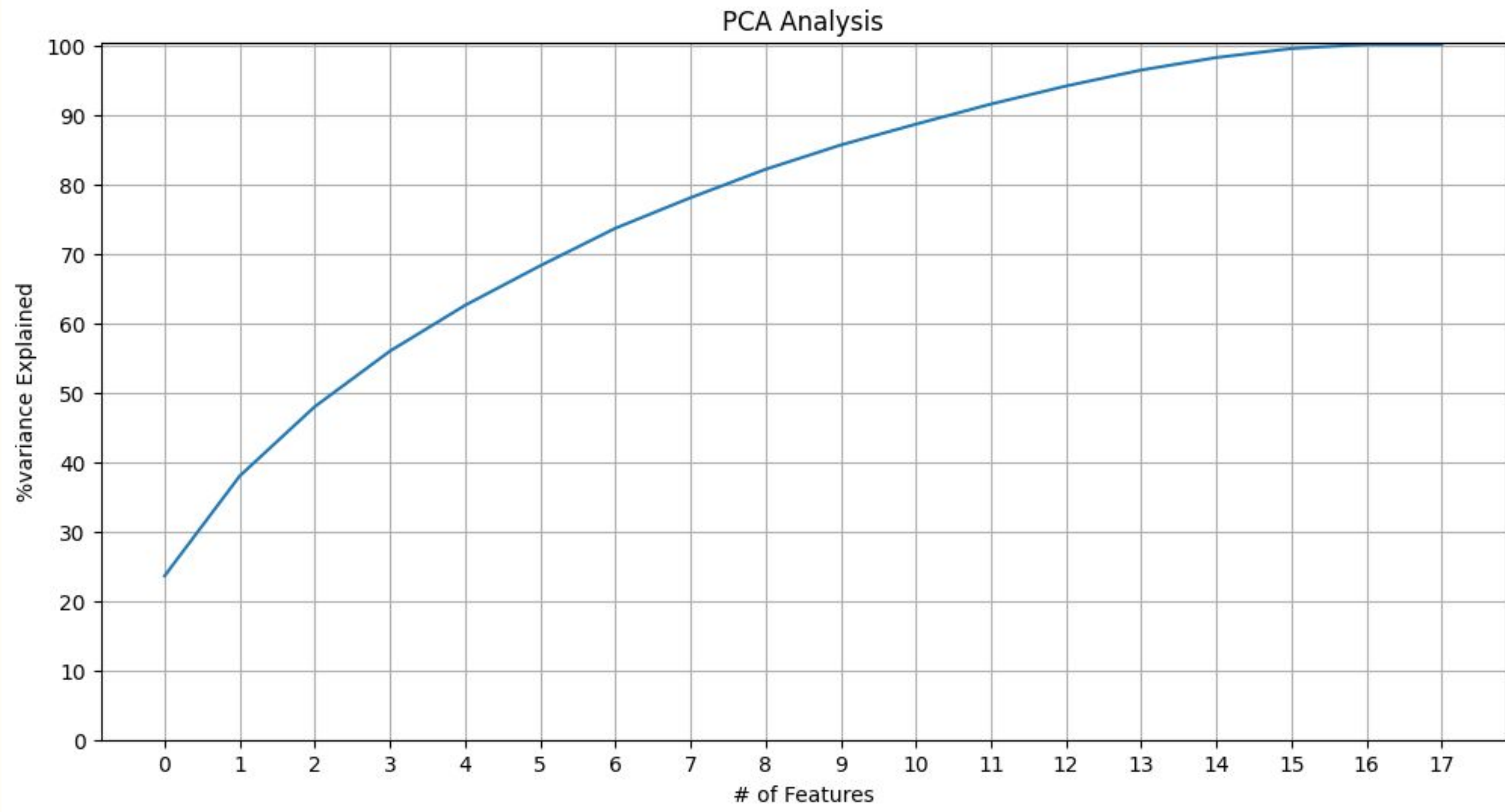
```
➡ Indexes with infinite values in 'AVG_TRX_AMOUNT': Index([1458, 3590, 4497], dtype='int64')
```

It doesn't make sense that there is a purchase but no recorded transaction. The three inf values are dropped instead.

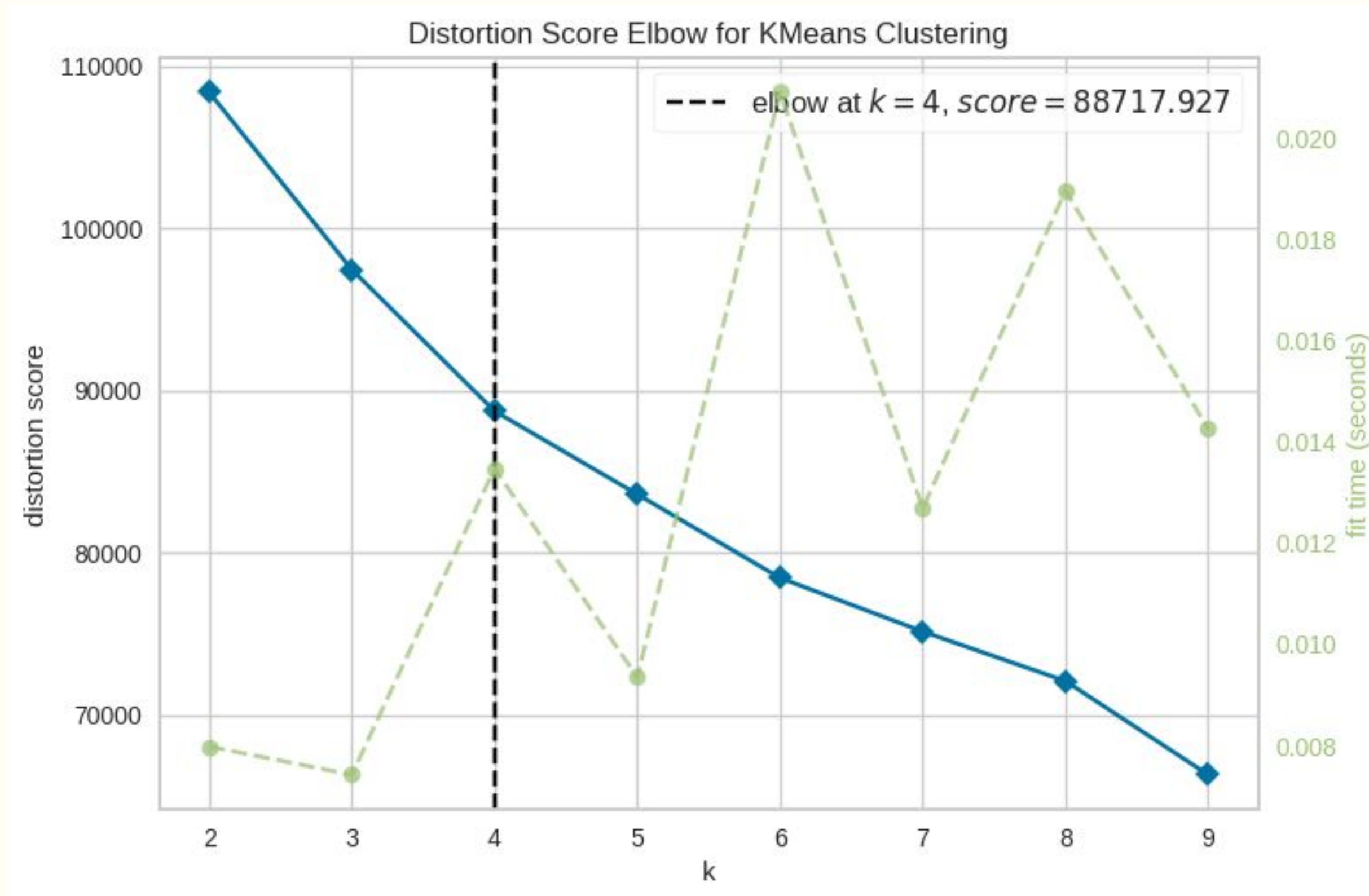
Correlation Matrix



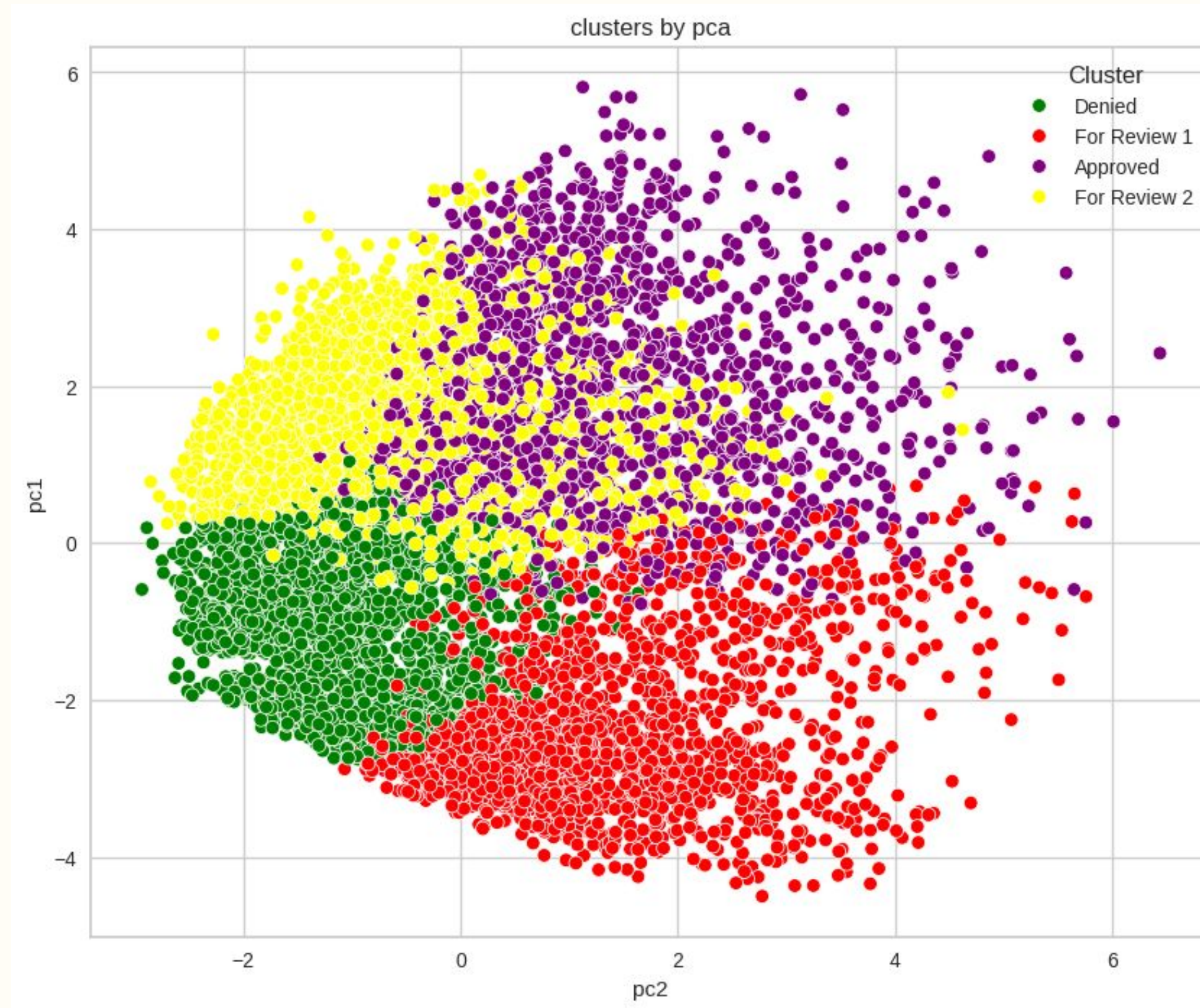
PCA Analysis



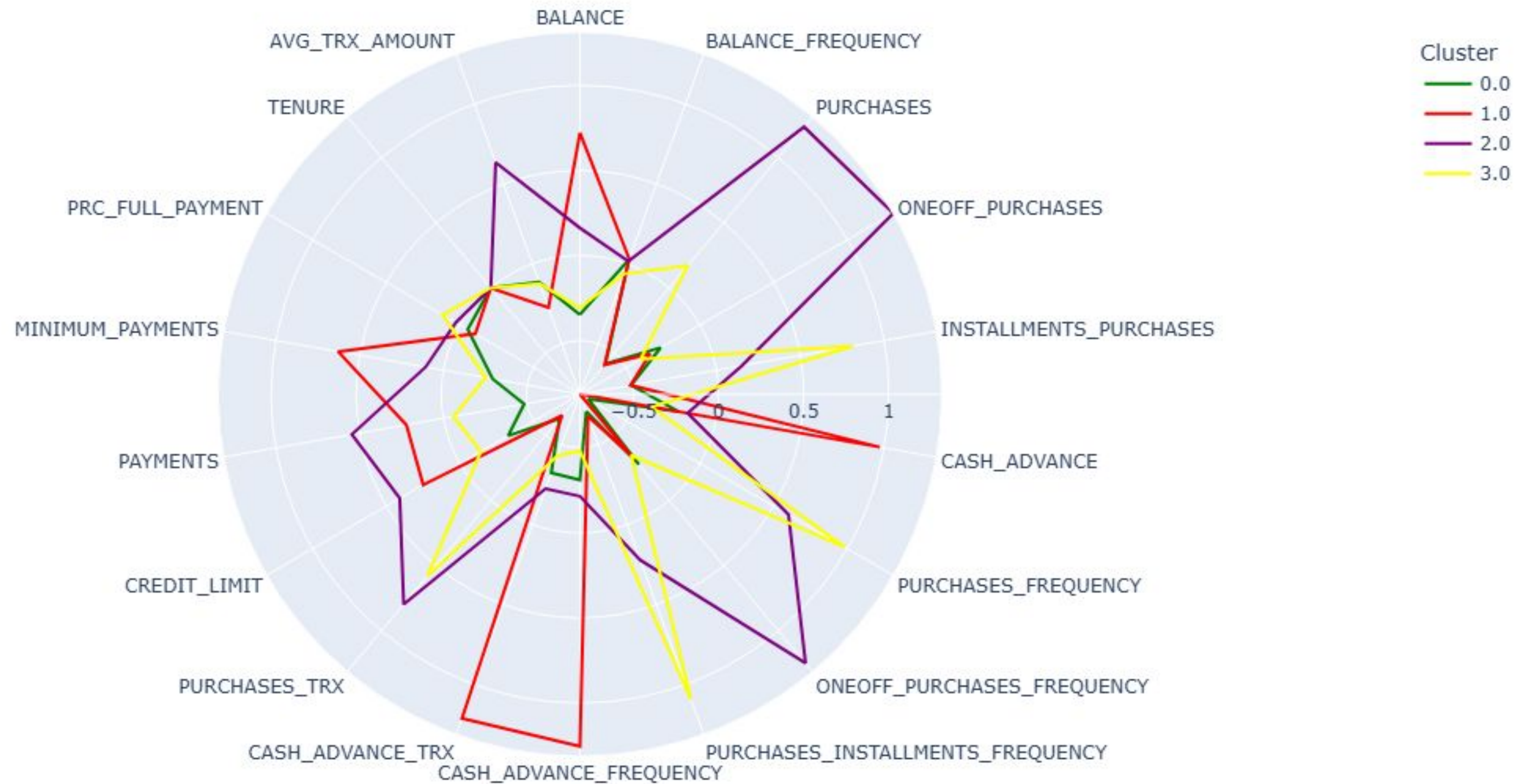
Distortion Score Elbow for KMeans Clustering



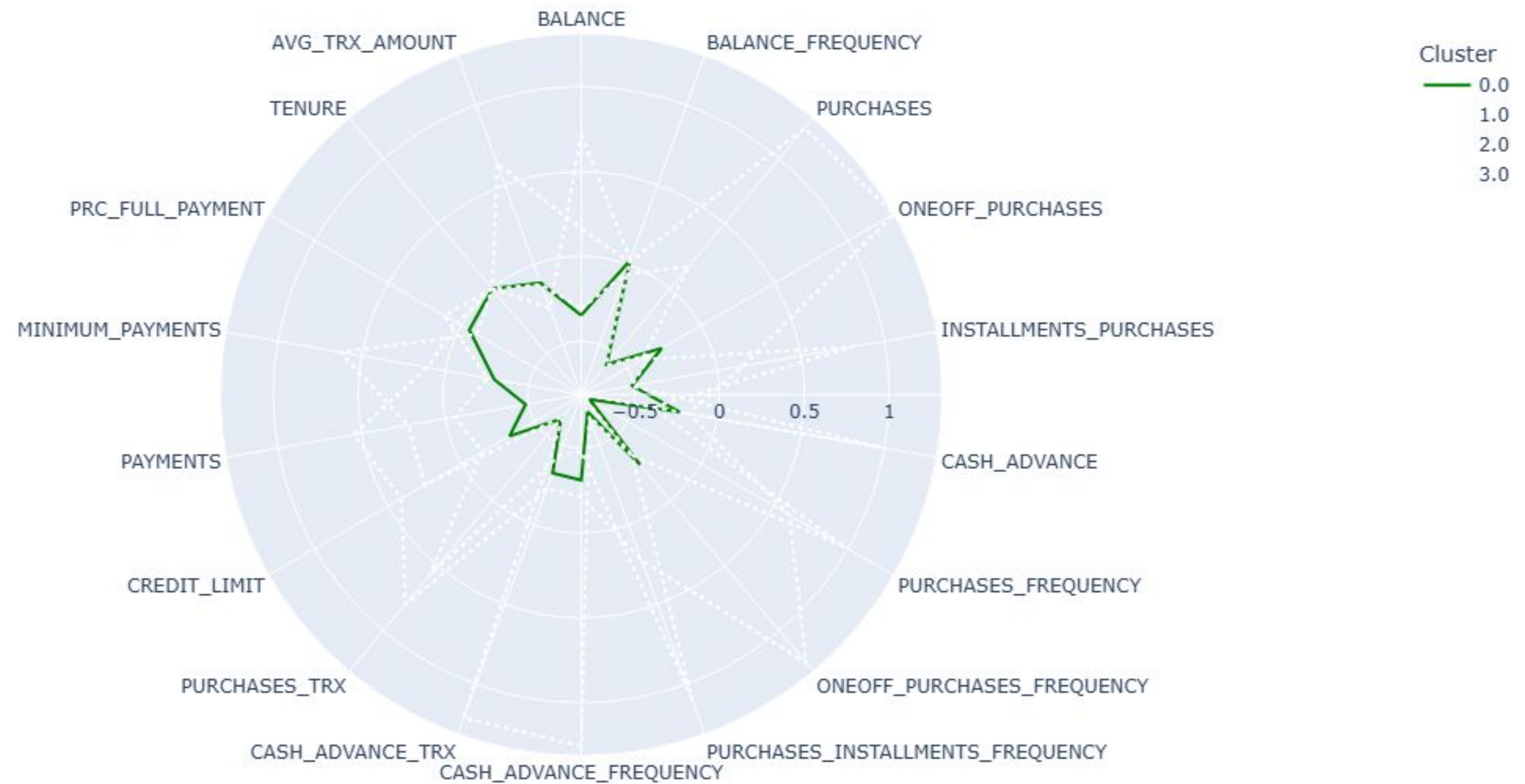
Clusters by PCA



Polar line graph

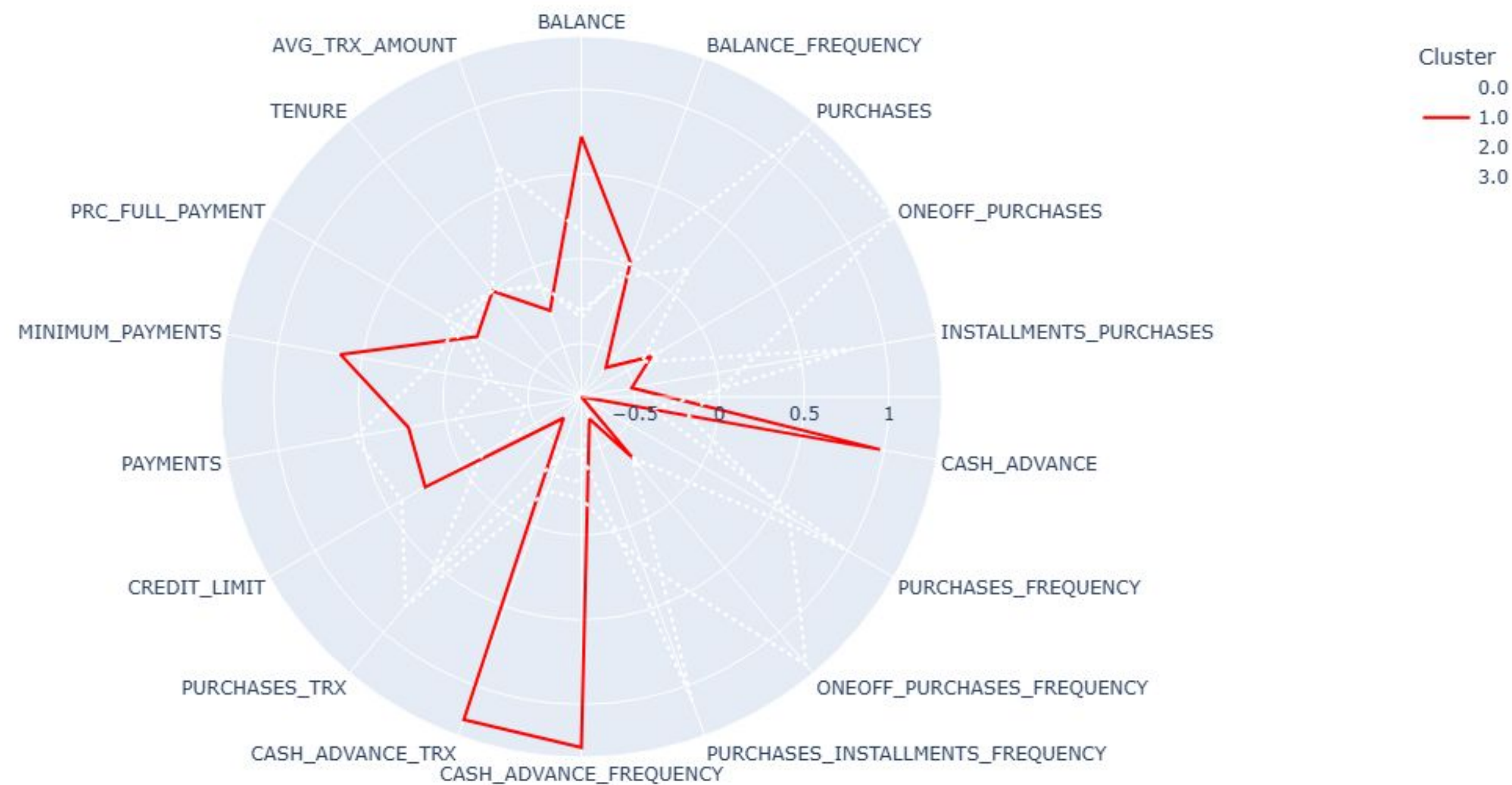


Polar line graph



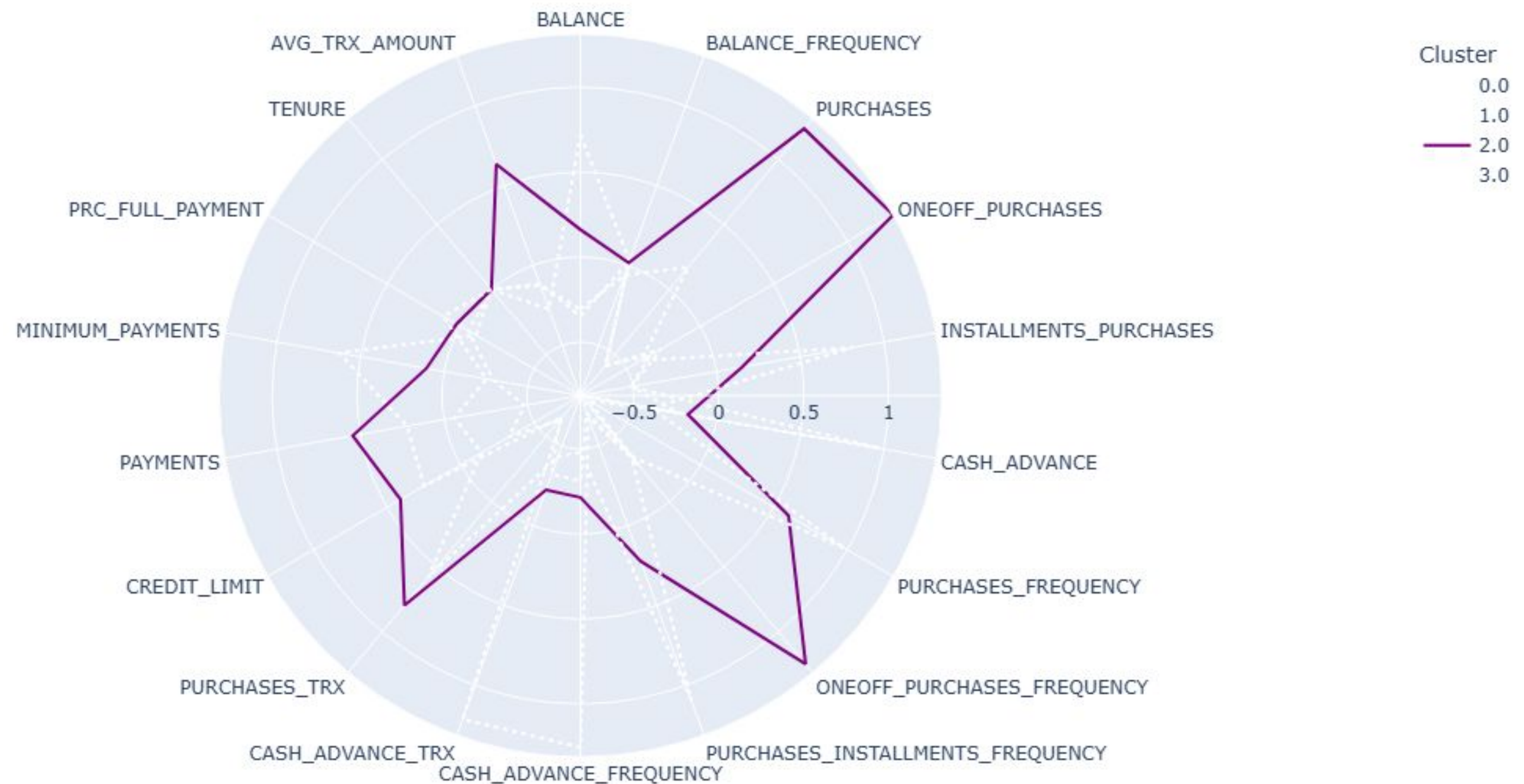
Cluster 0 - Low Activity Customers, Rarely use their Credit Cards. [Auto-Deny]

Polar line graph



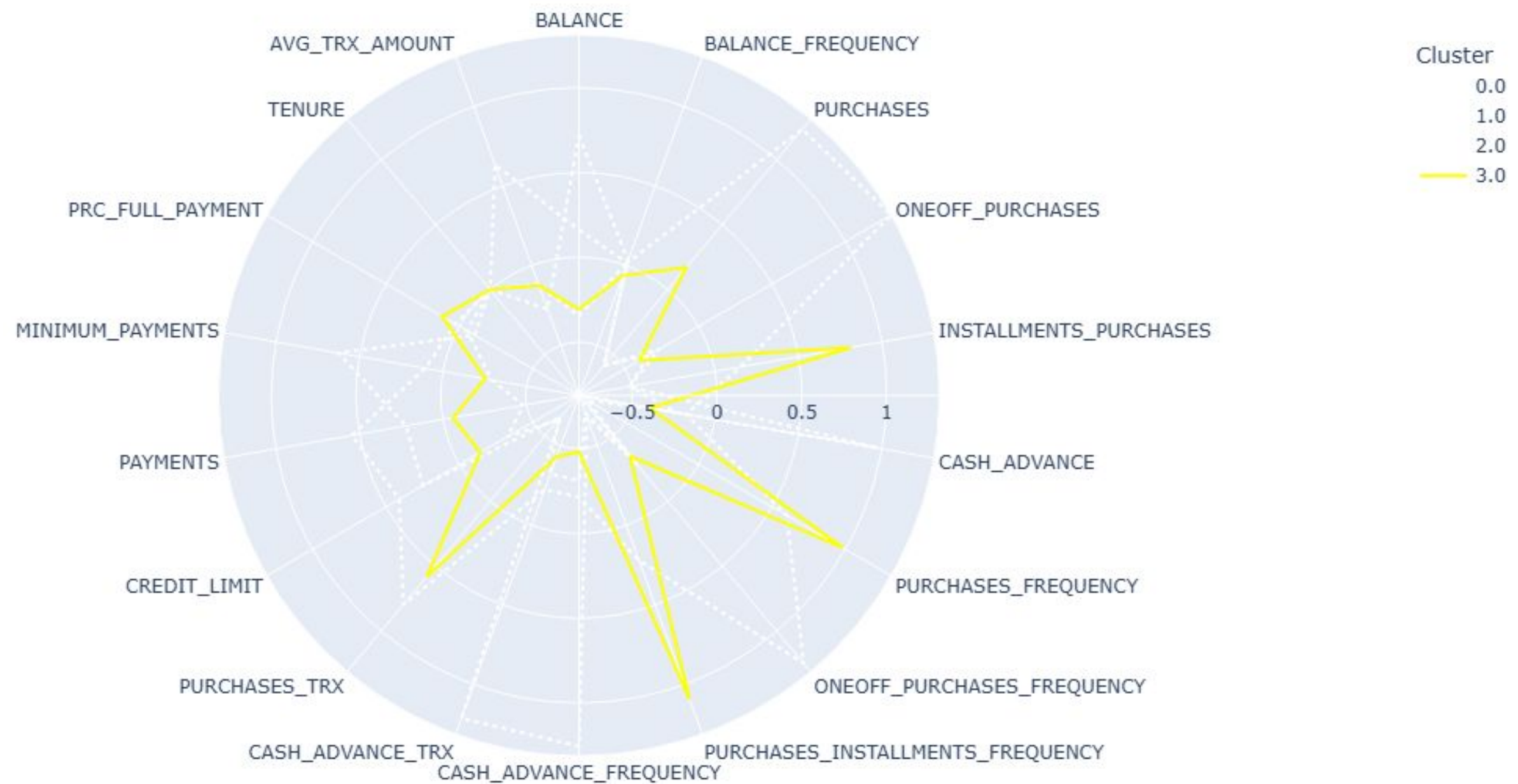
Cluster 1 - Very high use of cash advances, especially frequent transactions, low Repayment. [For Manual Review]

Polar line graph



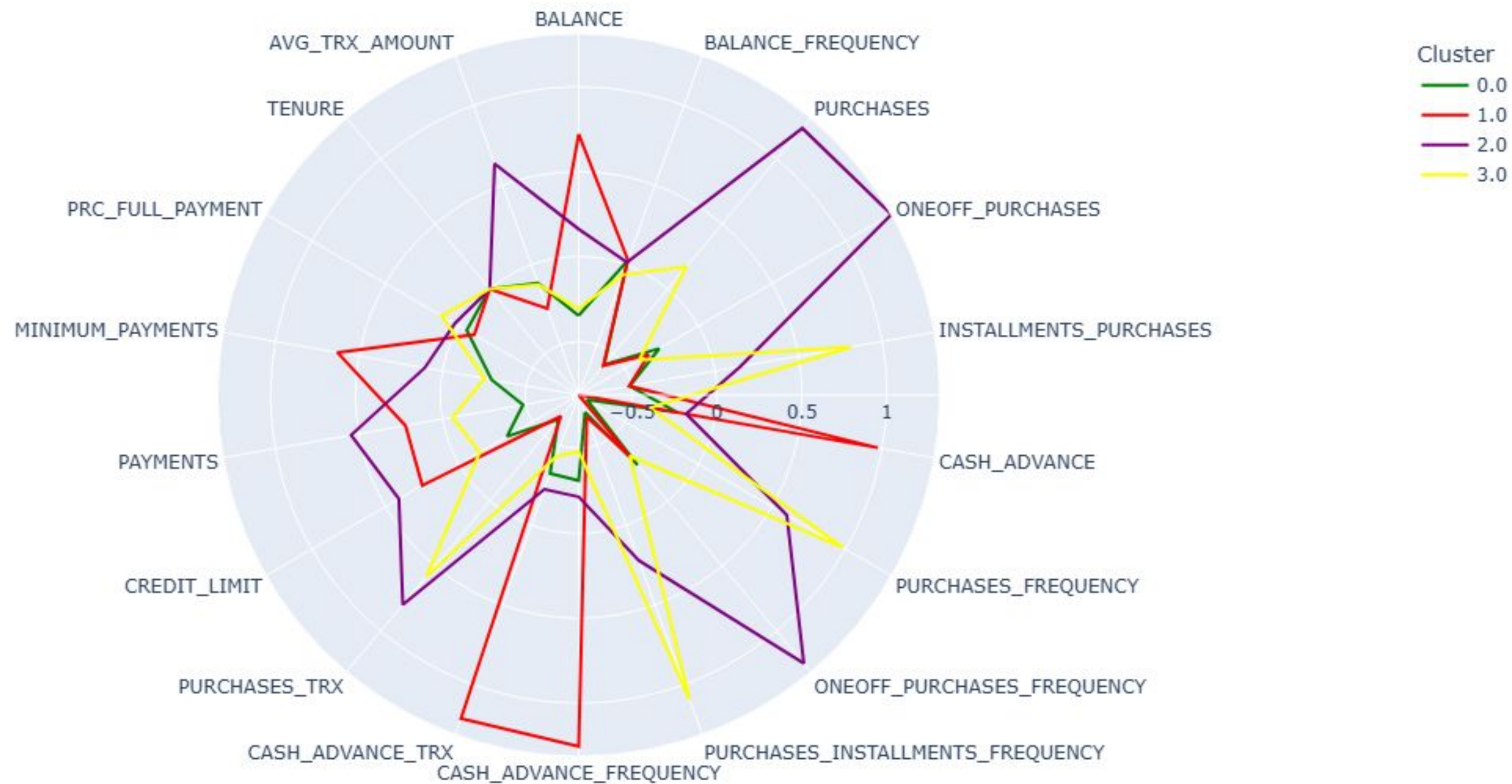
Cluster 2 - High levels of purchases (especially one-off), high payments. They spend and are good payors. [Auto-Approved]

Polar line graph



Cluster 3 - Preferred installment-based purchases, weaker payment metrics. [For Manual Review]

Polar line graph

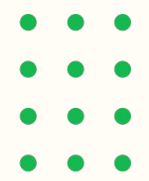


Confusion Matrix



Model Performance

Accuracy: 0.9720670391061452					
	precision	recall	f1-score	support	
Approved	0.94	0.97	0.95	350	
Denied	0.99	0.97	0.98	518	
For Review 1	0.98	0.97	0.97	402	
For Review 2	0.97	0.98	0.98	520	
accuracy			0.97	1790	
macro avg	0.97	0.97	0.97	1790	
weighted avg	0.97	0.97	0.97	1790	



Algorithm to assess Credit Application

LetMeCheck is BDI's machine-learning approval engine that evaluates a credit application by ingesting each applicant's verified attributes. It instantly routes decisions into three buckets:

- Approve – Low risk, meets all credit criteria
- Decline – High risk, low returns
- Review – Borderline cases needing human attention

BALANCE:	0
BALANCE F...	0
PURCHASES:	0
PURCHASE...	0
PURCHASE...	0
ONEOFF PU...	0
ONEOFF PU...	0
INSTALLME...	0
INSTALLME...	0
CASH ADVA...	1000
CASH ADVA...	1
CASH ADVA...	10
CREDIT LIM...	500
PAYMENTS:	0
MINIMUM P...	0
PERCENT F...	0
TENURE:	0

Assess

Predicted Cluster: Denied

**“THANK YOU
VERY MUCH”**

